Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/03/2026
PI Name:	Diaz Artiles, Ana Ph.D.		
Project Title:	Effects of Altered-Gravity on Perception and Bi-manual Coordination: Impacts on Functional Performance		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-0001	**Congressional District:**	17
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	08/01/2020	**End Date:**	09/30/2026
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
Contact Email:	becky.brocato@nasa.gov		
Flight Program:	Parabolic		
Flight Assignment:	NOTE: End date changed to 9/30/2026 per V. Lehman/NASA JSC (Ed., 10/8/25) NOTE: End date changed to 7/31/2025 per NSSC information (Ed., 9/22/24) NOTE: End date changed to 7/31/2024 per NSSC information (Ed., 6/15/22) NOTE: End date changed to 7/31/2022 per NSSC information (Ed., 7/6/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dunbar, Bonnie Ph.D. (Texas A&M University) Kennedy, Deanna Ph.D. (Texas A&M University)		
Grant/Contract No.:	80NSSC20K1499		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Many of the activities associated with spaceflight require individuals to use both limbs simultaneously to accomplish the task. Motor control, as well as visual performance and spatial orientation are disrupted by gravitational transitions between 1 G and 0 G, but very little is known about the sensorimotor deficits between 0 G and 1 G. The objective of this analog-based research effort is to investigate the impact of partial G-levels on bimanual coordination tasks that are operationally relevant for spaceflight. The same set of human subjects will participate in two different bimanual coordination tasks during parabolic flight, which will deliver G-levels of 0, 0.25, 0.5, 0.75, 1, and 1.8 G. Sensorimotor dose-response curves will be generated between bimanual coordination operational variables as a function of G-level, and G-thresholds (which indicate when performance decrements occur) will be determined. We will also quantify the risk associated with the use of a common motion sickness drug (promethazine) during bimanual coordination tasks. Results will provide critical information for current and future countermeasure development and in-flight prescriptions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project investigates the influence of gravity on bimanual coordination using a variety of altered-gravity analogs. Results will provide critical information for current and future sensorimotor-related countermeasures and in-flight prescription. In addition, this research effort has direct application to bimanual coordination tasks on Earth, for example during complex tasks that require a coordinated two-limb movement, as well as for rehabilitation purposes.
Task Progress:	We successfully conducted our parabolic flight campaign in June 2023, and we have been focusing on data analysis, dissemination, and publication activities since then. During the current reporting period, four peer-reviewed manuscripts were published, with additional analyses and manuscripts still in preparation. The published studies examined: (1) the neurophysiological effects of partial gravity on bimanual control, (2) gravity as a contextual control parameter in coordination dynamics, (3) the effects of promethazine on motion sickness during altered-gravity exposure, and (4) muscle activation coupling during bimanual tasks performed in microgravity and partial gravity. The publications are listed below: Kennedy DM, Neto OP, Weinrich M, Abbott R, Keller N, Rao R, Wright TJ, Dunbar BJ, Diaz-Artiles A. “Neurophysiological effects of partial gravity on bimanual control: A parabolic flight study.” npj Microgravity 2026 Apr. doi: 10.1038/s41526-026-00595-z. Weinrich M, Neto OP, Rao R, Abbott R, Keller N, Wright TJ, Dunbar BJ, Diaz-Artiles A, Kennedy DM. “Gravity as a contextual control parameter in coordination dynamics: Phase-specific stability during parabolic flight.” Hum Mov Sci. 2026 Apr. doi: 10.1016/j.humov.2026.103468. Abbott R, Weinrich M, Keller N, Wright TJ, Dunbar BJ, Denise P, Kennedy DM, Diaz-Artiles A. “Promethazine effects on motion sickness during altered gravity induced by parabolic flight”. Aerosp Med Hum Perform. 2025 Nov;96(11):976-84. doi: 10.3357/AMHP.6669.2025. Neto OP, Weinrich M, Abbott R, Keller N, Rao R, Wright TJ, Dunbar BJ, Diaz-Artiles A, Kennedy, DM. “Reduced gravity weakens muscle activation coupling and destabilizes bimanual force control.” J Neurophysiol. 2026 Apr;41941338. doi: 10.1152/jn.00632.2025. We also continue to investigate bimanual coordination disruption and adaptation following gravitational transitions experienced during parabolic flight, and an additional manuscript is currently in preparation. Collectively, our findings demonstrate that altered-gravity environments significantly influence bimanual coordination performance and the underlying neuromuscular processes that support motor control. Across studies, reductions in gravitational loading were associated with decreases in force output and movement harmonicity, increases in timing variability, and reductions in muscle activation coupling. These effects were generally more pronounced during the more complex coordination patterns. Analyses of electromyographic activity further showed that reduced gravity weakened intermuscular coupling across multiple frequency bands, particularly in microgravity, supporting the view that gravitational loading is an important contextual factor in sensorimotor control. The coordination dynamics analyses further demonstrated that reduced gravitational loading decreased the stability of coordination patterns. Performance was generally most stable at Earth gravity and became progressively more variable as gravitational loading decreased, particularly for coordination patterns that are inherently more difficult to perform. Partial-gravity conditions frequently produced intermediate responses between Earth gravity and microgravity, providing additional insight into gravitational dose-response relationships for sensorimotor performance. Motion sickness was assessed throughout the campaign using subjective measures collected before, during, and after flight. Participants receiving a placebo generally reported greater symptom severity than those receiving promethazine, particularly during microgravity exposure. These findings support the effectiveness of promethazine as a countermeasure for motion sickness during parabolic flight operations. During the current reporting period, we also initiated the student augmentation project examining lower body negative pressure (LBNP) as a ground-based analogue for gravitational transitions. This effort aims to characterize both bimanual coordination performance and cardiovascular responses during simulated gravitational transitions, and to compare these responses with those observed during parabolic flight. Data collection and analysis are ongoing. Overall, the project has generated a unique dataset spanning multiple gravity levels, coordination tasks, and pharmacological conditions. The resulting findings advance our understanding of the effects of altered gravity on functional sensorimotor performance and provide new insights relevant to future lunar, Martian, and deep-space exploration missions.
Bibliography Type:	Description: (Last Updated: 07/10/2026)
Abstracts for Journals and Proceedings	Rao R, Weinrich M, Abbott R, Neto OP, Kennedy DM, Diaz-Artiles A. "Lower-body negative pressure as a gravitational transition analogue for bimanual coordination testing." 2026 NASA Human Research Program Investigators’ Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators’ Workshop, Virtual, April 7-9, 2026. , Apr-2026

Articles in Peer-reviewed Journals	Abbott R, Weinrich MM, Keller NW, Wright TJ, Dunbar BJ, Denise P, Kennedy DM, Diaz-Artiles A. "Promethazine effects on motion sickness during altered gravity induced by parabolic flight." Aerosp Med Hum Perform. 2025 Nov 1;96(11):976-84. http://dx.doi.org/10.3357/AMHP.6669.2025 ; PMID: 41253367 , Nov-2025
Articles in Peer-reviewed Journals	Neto OP, Weinrich M, Abbott R, Keller N, Rao R, Wright TJ, Dunbar BJ, Diaz-Artiles A, Kennedy, DM. "Reduced gravity weakens muscle activation coupling and destabilizes bimanual force control." J Neurophysiol. 2026 May 1;135(5):1246-1259. https://doi.org/10.1152/jn.00632.2025 ; PubMed PMID: 41941338 , May-2026
Articles in Peer-reviewed Journals	Kennedy DM, Neto OP, Weinrich M, Abbott R, Keller N, Rao R, Wright TJ, Dunbar BJ, Diaz-Artiles A. "Neurophysiological effects of partial gravity on bimanual control: A parabolic flight study." npj Microgravity. 2026 Apr 16;12(1):53. https://doi.org/10.1038/s41526-026-00595-z ; PubMed PMID: 41991536; PubMed Central PMCID: PMC13272968 , Apr-2026
Articles in Peer-reviewed Journals	Weinrich M, Neto OP, Rao R, Abbott R, Keller N, Wright TJ, Dunbar BJ, Diaz-Artiles A, Kennedy DM. "Gravity as a contextual control parameter in coordination dynamics: Phase-specific stability during parabolic flight." Hum Mov Sci. 2026 Apr:106:103468. https://doi.org/10.1016/j.humov.2026.103468 ; PubMed PMID: 41747503 , Apr-2026
Dissertations and Theses	Weinrich M. "Lissajous feedback utilization in Parkinson's disease: A bimanual coordination perspective." Ph.D. Dissertation, Kinesiology, Texas A&M University, February 2026. , Feb-2026